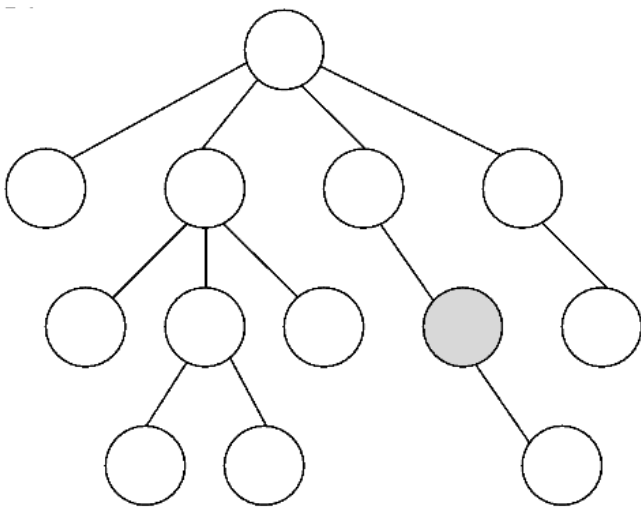


Q3. Same Degree of Relative

The algorithm of relative degree starts by yourself, counts up to the common ancestor of that person, and counts down to that person. The total number of counts is the degree of relative. For example, the picture below is a relative diagram of a family, and each member of the family is represented by a number code. Members 1~4 are children of member 0; members 5~7 are children of member 2, and so on. Member 8 and member 1 are 3-degree relative, member 8 and member 10 are 5-degree. Given a member A, please find all members in the family whose relative degree is B, and output according to the member code from small to large. In the figure below, all members whose 4-degree relatives of member 8 are 5, 6, 7, 9.



Input:

The first line has three numbers, N, A and B. N represents the total numbers of family members, A represents the family member who wants to find the relative, and B represents the degree of relative will be found.

The 2~N lines, each consists of two numbers, P and Q. Means that Q is the child of P.

Output:

Output consists of multiple numbers, represents all family members of A whose degree is equal to B. Sorting the output increasing.

Sample input:

```
13 8 4
0 1
0 2
0 3
```

0 4

2 5

2 6

2 7

3 8

4 9

6 10

6 11

8 12

Sample output:

5 6 7 9